Chapter 30

Campbell Biology, AP Edition, 12e

Fireweed (Chamerion angustifolium) on Mount St. Helens.

Chapter 30

Plant Diversity II: The Evolution of Seed Plants

Fireweed seed.

Figure 30.1

In 1980, Mount St. Helens in Washington state erupted, leaving the region covered in ash and devoid of visible life. Just a few years later, seed plants such as fireweed (*Chamerion angustifolium*) had already colonized the barren landscape—an example of seed plants' versatility and success as the dominant producers on land.

What adaptations have enabled seed plants to make up the vast majority of plant biodiversity?

Reduced male and female gametophytes develop within parental sporophytes, protected from environmental stresses. In gymnosperms, the gametophytes develop within cones. In angiosperms, the gametophytes develop within flowers. Pollen protects male gametophytes (which produce sperm) and can be transported by wind or animals. Ovules protect female gametophytes (which produce eggs). An ovule fertilized by pollen develops into a seed.

Key Concepts

- **30.1Seeds and pollen grains are key adaptations for life on land****
- **30.2Gymnosperms bear "naked" seeds, typically on cones****
- **30.3The reproductive adaptations of angiosperms include flowers and fruits****
- **30.4Human welfare depends on seed plants****

Advanced placement edition

Flowering plants regulate the development of pollen and seeds, responding to environmental conditions such as water availability and temperature **(BIG IDEA 2)** in ways that optimize the survival of the next generation **(BIG IDEA 4)**.

Study Tip

Make flowcharts: Make simple flowcharts showing the main events in the life cycles of a pine and flowering plant. The example shown is the beginning of a flowchart summarizing the pine life cycle.

A flow chart. Main events in pine life cycle. First. Pollen grain is released from a pollen cone. Second. Pollen grain reaches an ovule and forms a pollen tube.

Fireweed and other early arrivals reached the blast zone in Figure 30.1 as seeds. A **seed** consists of an embryo and its food supply, surrounded by a protective coat. When mature, seeds are dispersed from their parent by wind or other means, enabling them to colonize distant locations.

Plants not only have affected the recovery of regions such as Mount St. Helens but also have transformed Earth. Continuing the saga of how this occurred, this chapter follows the emergence and diversification of the group to which fireweed belongs, the seed plants.

Get Ready for This Chapter

In this chapter, you will learn about the general characteristics and evolutionary history of seed plants, as well as their impact on human society. Before beginning this chapter, you should be able to do the following:

- Distinguish between analogy, homology, and descent with modification (see Concept 22.3).
- Describe a life cycle that exhibits alternation of generations (see Figure 13.6b and Figure 29.5).
- Define the terms *gametophyte* and *sporophyte*, explaining how each contributes to sexual reproduction in plants (see Figure 29.5).

The following questions provide a quick check of your basic knowledge in these areas.

1. What is the process by which a trait present in an ancestral organism is modified by natural selection over time in descendants of that ancestor?

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analogy
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That's incorrect. Try again.

Incorrect. Review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered (by natural selection) in its descendants over time as they face different environmental conditions. As a result, related species can have characteristics that have an underlying similarity yet function differently. An underlying similarity resulting from common ancestry is known as homology, while similarity due to convergent evolution is known as analogy.

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convergent evolution
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That's incorrect. Try again.

Incorrect. Review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered (by natural selection) in its descendants over time as they face different environmental conditions. As a result, related species can have characteristics that have an underlying similarity yet function differently. An underlying similarity resulting from common ancestry is known as homology, while similarity due to convergent evolution is known as analogy.

```
descent with modification
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That's correct.

Correct! If you had trouble with this question, review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered (by natural selection) in its descendants over time as they face different environmental conditions. As a result, related species can have characteristics that have an underlying similarity yet function differently. An underlying similarity resulting from common ancestry is known as homology, while similarity due to convergent evolution is known as analogy.

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homology
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That's incorrect. Try again.

Incorrect. Review the following material: Evolution is a process of descent with modification: Characteristics present in an ancestral organism are altered (by natural selection) in its descendants over time as they face different environmental conditions. As a result, related species can have characteristics that have an underlying similarity yet function differently. An underlying similarity resulting from common ancestry is known as homology, while similarity due to convergent evolution is known as analogy.

2. A life cycle that exhibits alternation of generations includes the following:

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haploid multicellular organisms only
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That's incorrect. Try again.

Incorrect. Review the following material: The life cycles of all plants alternate between two generations of distinct multicellular organisms: multicellular haploid gametophytes and multicellular diploid sporophytes. Each generation gives rise to the other. That is why this process is called alternation of generations.

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multicellular haploid organisms and a diploid
unicellular stage
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That's incorrect. Try again.

Incorrect. Review the following material: The life cycles of all plants alternate between two generations of distinct multicellular organisms: multicellular haploid gametophytes and multicellular diploid sporophytes. Each generation gives rise to the other. That is why this process is called alternation of generations.

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a unicellular haploid stage and diploid multicellular
organisms
```

That's incorrect. Try again.

Incorrect. Review the following material: The life cycles of all plants alternate between two generations of distinct multicellular organisms: multicellular haploid gametophytes and multicellular diploid sporophytes. Each generation gives rise to the other. That is why this process is called alternation of generations.

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multicellular haploid organisms and multicellular
diploid organisms
```

That's correct.

Correct! If you had trouble with this question, review the following material: The life cycles of all plants alternate between two generations of distinct multicellular organisms: multicellular haploid gametophytes and multicellular diploid sporophytes. Each generation gives rise to the other. That is why this process is called alternation of generations.

3. In plants, the _____ produces haploid gametes (eggs and sperm) by mitosis, whereas the _____ produces haploid spores by meiosis.

```
gametophyte; sporophyte
```

That's correct.

Correct! If you had trouble with this question, review the following material: In plants, the multicellular haploid gametophyte ("gamete-producing plant") is named for its production by mitosis of haploid gametes—eggs and sperm—that fuse during fertilization, forming diploid zygotes. Mitotic division of the zygote produces a multicellular diploid sporophyte ("spore-producing plant"). Meiosis in a mature sporophytes produces haploid spores, reproductive cells that can develop into a new haploid organism without fusing with another cell. Mitotic division of the spore cell produces a new multicellular gametophyte, and the cycle begins again.

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sporophyte; gametophyte
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That's incorrect. Try again.

Incorrect. Review the following material: In plants, the multicellular haploid gametophyte ("gamete-producing plant") is named for its production by mitosis of haploid gametes—eggs and sperm—that fuse during fertilization, forming diploid zygotes. Mitotic division of the zygote produces a multicellular diploid sporophyte ("spore-producing plant"). Meiosis in a mature sporophytes produces haploid spores, reproductive cells that can develop into a new haploid organism without fusing with another cell. Mitotic division of the spore cell produces a new multicellular gametophyte, and the cycle begins again.

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zygote; embryo
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That's incorrect. Try again.

Incorrect. Review the following material: In plants, the multicellular haploid gametophyte ("gamete-producing plant") is named for its production by mitosis of haploid gametes—eggs and sperm—that fuse during fertilization, forming diploid zygotes. Mitotic division of the zygote produces a multicellular diploid sporophyte ("spore-producing plant"). Meiosis in a mature sporophytes produces haploid spores, reproductive cells that can develop into a new haploid organism without fusing with another cell. Mitotic division of the spore cell produces a new multicellular gametophyte, and the cycle begins again.

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embryo; zygote
```

That's incorrect. Try again.

Incorrect. Review the following material: In plants, the multicellular haploid gametophyte ("gamete-producing plant") is named for its production by mitosis of haploid gametes—eggs and sperm—that fuse during fertilization, forming diploid zygotes. Mitotic division of the zygote produces a multicellular diploid sporophyte ("spore-producing plant"). Meiosis in a mature sporophytes produces haploid spores, reproductive cells that can develop into a new haploid organism without fusing with another cell. Mitotic division of the spore cell produces a new multicellular gametophyte, and the cycle begins again.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 30.1Explain how seeds and pollen grains were key adaptations for plants.
- 30.2Characterize gymnosperms and their life cycles.
- 30.3Describe the structure and function of flowers and fruits.

Concept 30.1: Seeds and pollen grains are key adaptations for life on land

We begin with an overview of terrestrial adaptations that seed plants added to those already present in nonvascular plants (bryophytes) and seedless vascular plants (see Concept 29.1). In addition to seeds, all seed plants have reduced gametophytes, heterospory, ovules, and pollen. As we'll see, these adaptations helped seed plants cope with conditions such as drought and exposure to ultraviolet (UV) radiation in sunlight. They also freed seed plants from requiring water for fertilization, enabling reproduction under a broader range of conditions than in seedless plants.

Advantages of Reduced Gametophytes

Mosses and other bryophytes have life cycles dominated by gametophytes, whereas ferns and other seedless vascular plants have sporophyte-dominated life cycles. The evolutionary trend of gametophyte reduction continued further in the vascular plant lineage that led to seed plants **(Figure 30.2)**. While the gametophytes of seedless vascular plants are visible to the naked eye, the gametophytes of most seed plants are microscopic.

Figure 30.2

Gametophyte-sporophyte relationships in different plant groups.

A table illustrates gametophyte, sporophyte, and examples for the following plant groups: mosses, ferns, and seed plants.

The table is as follows:

The table is as follows:
blank
Gametophyte
Sporophyte
Example

MAKE CONNECTIONS *In seed plants, how might retaining the gametophyte within the sporophyte affect embryo fitness? (See*Concepts 17.5,***23.1**, and***23.4***to review mutagens, mutations, and fitness.)*

*For suggested answer, see*Appendix A.**

This miniaturization allowed for an important evolutionary innovation in seed plants: Their tiny gametophytes can develop from spores retained within the sporangia of the parental sporophyte. This arrangement can protect the gametophytes from environmental stresses. For example, the moist reproductive tissues of the sporophyte shield the gametophytes from UV radiation and protect them from drying out. This relationship also enables the developing gametophytes to obtain nutrients from the parental sporophyte. In contrast, the free-living gametophytes of seedless vascular plants must fend for themselves.

Heterospory: The Rule Among Seed Plants

You read in Concept 29.3 that most seedless plants are *homosporous*—they produce one kind of spore, which usually gives rise to a bisexual gametophyte. Ferns and other close relatives of seed plants are homosporous, suggesting that seed plants had homosporous ancestors. At some point, seed plants or their ancestors became *heterosporous*, producing two kinds of spores: Megasporangia on modified leaves called megasporophylls produce *megaspores* that give rise to female gametophytes, and microsporangia on modified leaves called microsporophylls produce *microspores* that give rise to male gametophytes. Each megasporangium has one megaspore, whereas each microsporangium has many microspores.

As noted previously, the miniaturization of seed plant gametophytes probably contributed to the great success of this clade. Next, we'll look at the development of the female gametophyte within an ovule and the development of the male gametophyte in a pollen grain. Then we'll follow the transformation of a fertilized ovule into a seed.

Ovules and Production of Eggs

Although a few species of seedless plants are heterosporous, seed plants are unique in retaining the megasporangium within the parent sporophyte. A layer of sporophyte tissue called **integument** envelops and protects the megasporangium. Gymnosperm megasporangia are surrounded by one integument, whereas those in angiosperms usually have two integuments. The whole structure—megasporangium, megaspore, and their integument(s)—is called an **ovule(Figure 30.3a)**. Inside each ovule (from the Latin *ovulum*, little egg), a female gametophyte develops from a megaspore and produces one or more eggs.

Figure 30.3

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From ovule to seed in a gymnosperm.
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A diagram of a, unfertilized ovule, b, fertilized ovule, and c, gymnosperm seed.

Parts are as follows.

a Unfertilized ovule. In this longitudinal section through the ovule of a pine (a gymnosperm), a fleshy megasporangium is surrounded by a protective layer of tissue called an integument. The micropyle, the only opening through the integument, allows entry of a pollen grain.

b Fertilized ovule. A megaspore develops into a female gametophyte, which produces an egg. The pollen grain, which had entered through the micropyle, contains a male gametophyte. The male gametophyte develops a pollen tube that discharges sperm, thereby fertilizing the egg.

c Gymnosperm seed. Fertilization initiates the transformation of the ovule into a seed, which consists of a sporophyte embryo, a food supply, and a protective seed coat derived from the integument. The megasporangium dries out and collapses.

Haploid Cells: Megaspore (n), Pollen grain (n), Female gametophyte (n), Egg nucleus (n), Discharged sperm nucleus (n), Male gametophyte (within a germinated pollen grain) (n), Food supply (female gametophyte tissue) (n). Diploid Cells: Integument (2 n), Megasporangium (2 n), Embryo (2 n) (new sporophyte).

VISUAL SKILLS *Based on this figure, a gymnosperm seed contains cells from how many different plant generations? Identify the cells and whether each is haploid or diploid.*

*For suggested answer, see*Appendix A.**

Figure 30.3 Walkthrough

From ovule to seed in a gymnosperm

Watch an explanation of this figure in the Figure 30.3 Walkthrough: From ovule to seed in a gymnosperm.

image

Play

image

Pollen and Production of Sperm

A microspore develops into a **pollen grain** that consists of a male gametophyte enclosed within the pollen wall. (The wall's outer layer is made of molecules secreted by sporophyte cells, so we refer to the male gametophyte as being *in* the pollen grain, not *equivalent to* the pollen grain.) Sporopollenin in the pollen wall protects the pollen grain as it is transported by wind or by hitchhiking on an animal. The transfer of pollen to the part of a seed plant that contains the ovules is called **pollination**. If a pollen grain germinates (begins growing), it gives rise to a pollen tube that discharges sperm into the female gametophyte within the ovule, as shown in **Figure 30.3b**.

In nonvascular plants and seedless vascular plants such as ferns, free-living gametophytes release flagellated sperm that swim through a film of water to reach eggs. Given this requirement, it is not surprising that many of these species live in moist habitats. But a pollen grain can be carried by wind or animals, eliminating the dependence on water for sperm transport. The ability of seed plants to transfer sperm without water likely contributed to their colonization of dry habitats. The sperm of seed plants also do not require motility because they are carried to the eggs by pollen tubes. The sperm of some gymnosperm species (such as cycads and ginkgos, shown in Figure 30.7) retain the ancient flagellated condition, but flagella have been lost in the sperm of most gymnosperms and all angiosperms.

The Evolutionary Advantage of Seeds

If a sperm fertilizes an egg of a seed plant, the zygote grows into a sporophyte embryo. As shown in **Figure 30.3c**, the ovule develops into a seed: the embryo, with a food supply, packaged in a protective coat derived from the integument(s).

Seeds provide protection from harsh conditions and facilitate dispersal to new habitats—as do the spores of seedless plants. Moss spores, for example, may survive even if the local environment becomes too cold, too hot, or too dry for the mosses themselves to live. Their tiny size enables the spores to be dispersed in a dormant state to a new area, where they can germinate and give rise to new moss gametophytes if and when conditions are favorable enough for them to break dormancy. Spores were the main way that mosses, ferns, and other seedless plants spread over Earth for the first 100 million years of plant life on land.

Although mosses and other seedless plants continue to be very successful today, seeds represent a major evolutionary innovation that contributed to the opening of new ways of life for seed plants. What advantages do seeds provide over spores? Spores are usually single-celled, whereas seeds are multicellular, consisting of an embryo protected by a layer of tissue, the seed coat. A seed can remain dormant for days, months, or even years after being released from the parent plant, whereas most spores have shorter lifetimes. Also, unlike spores, seeds have a supply of stored food. Most seeds land close to their parent sporophyte plant, but some are carried long distances (up to hundreds of kilometers) by wind or animals. If conditions are favorable where it lands, the seed can emerge from dormancy and germinate, with its stored food providing critical support for growth as the sporophyte embryo emerges as a seedling. As we explore in the **Scientific Skills Exercise**, some seeds have germinated after more than 1,000 years.

Scientific Skills Exercise

Using Natural Logarithms to Interpret Data

APScience Practices 5, 6

How Long Can Seeds Remain Viable in Dormancy? Environmental conditions can vary greatly over time, and they may not be favorable for germination when seeds are produced. One way that plants cope with such variation is through seed dormancy. Under favorable conditions, seeds of some species can germinate after many years of dormancy.

One unusual opportunity to test how long seeds can remain viable occurred when seeds from date palm trees (*Phoenix dactylifera*) were discovered under the rubble of a 2,000-year-old fortress near the Dead Sea. As you saw in the Chapter 2 Scientific Skills Exercise and Concept 25.2, scientists use radiometric dating to estimate the ages of fossils and other old objects. In this exercise, you will estimate the ages of three of these ancient seeds by using natural logarithms.

A palm tree in a planter.

How the Experiment Was Done Scientists measured the fraction of carbon-14 that remained in three ancient date palm seeds: two that were not planted and one that was planted and germinated. For the germinated seed, the scientists used a seed coat fragment found clinging to a root of the seedling. (The seedling grew into the plant in the photo.)

Data from the Experiment This table shows the fraction of carbon-14 remaining from the three ancient date palm seeds.

A table lists the fractions of carbon-14 remaining in seeds 1, 2, and 3.

The table is as follows:

The table is as follows:
Blank
Seed 1 (not planted)
Seed 2 (not planted)
Seed 3 (germinated)

Data from S. Sallon et al., Germination, genetics, and growth of an ancient date seed, *Science* 320:1464 (2008).

INTERPRET THE DATA

A logarithm is the power to which a base is raised to produce a given number x. For example, if the base is 10 and x=100, the logarithm of 100 equals 2 (because 102=100). A natural logarithm (ln) is the logarithm of a number x to the base e, where e is about 2.718. Natural logarithms are useful in calculating rates of some natural processes, such as radioactive decay.

The equation F=e−kt describes the fraction F of an original isotope remaining after a period of t years; the exponent is negative because it refers to a *decrease* over time. The constant k provides a measure of how rapidly the original isotope decays. For the decay of carbon-14 to nitrogen-14, k=0.00012097.

To estimate t, the age of the three seeds, we rearrange the equation F=e−kt to find an equation for t:

t=−(ln Fk)

- Using the equation for t, the data from the table, and a calculator, estimate the ages of seed 1, seed 2, and seed 3.
- Why do you think there was more carbon-14 in the germinated seed?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Concept Check 30.1

- Contrast how sperm reach the eggs of seedless plants with how sperm reach the eggs of seed plants.
- What features not present in seedless plants have contributed to the success of seed plants on land?
- **WHAT IF?** If a seed could not enter dormancy, how might that affect the embryo's transport or survival?

*For suggested answers, see*Appendix A.**

Concept 30.2: Gymnosperms bear “naked” seeds, typically on cones

On an evolutionary tree, a branch goes to gymnosperms and angiosperms; gymnosperms are highlighted.

Branch point 1: First level-1 branch: Nonvascular plants (bryophytes). The second level-1 branch splits from branch point 2 into the first level-2 branch: seedless vascular plants. The second level-2 branch splits from branch point 3 into the first level-3 branch: Gymnosperms and the second level-3 branch: Angiosperms.

Extant seed plants form two sister clades: gymnosperms and angiosperms. Recall that gymnosperms have “naked” seeds exposed on sporophylls that usually form cones. (Angiosperm seeds are enclosed in chambers that mature into fruits.) Most gymnosperms are cone-bearing plants called **conifers**, such as pines, firs, and redwoods.

The Life Cycle of a Pine

As you read earlier, seed plant evolution has included three key reproductive adaptations: the miniaturization of their gametophytes; the advent of the seed as a resistant, dispersible stage in the life cycle; and the appearance of pollen as an airborne agent that brings gametes together. **Figure 30.4** shows how these adaptations come into play during the life cycle of a pine, a familiar conifer.

Figure 30.4

The life cycle of a pine.

Cycle of life is shown as 8 steps from ovulate and pollen cones through meiosis, fertilization, and the formation of a new embryo and seedling.

The cycle is as follows.

1. In most conifer species, each tree has both ovulate and pollen cones.
2. Microsporocytes (2 n) divide by meiosis, producing haploid microspores. A microspore develops into a pollen grain (a male gametophyte enclosed within the pollen wall).
3. An ovulate cone scale has two ovules, each containing a Megasporangium (2 n). Only one ovule is shown.

4. Pollination occurs when a pollen grain reaches the ovule. The pollen grain then germinates, forming a pollen tube that slowly digests its way through the Megasporangium (2 n).

5. While the pollen tube develops, the megasporocyte undergoes meiosis, producing four haploid cells. One survives as a megaspore (n).

6. The megaspore develops into a female gametophyte that contains two or three archegonia, each of which will form an egg.

7. By the time the eggs are mature, sperm cells have developed in the pollen tube, which extends to the female gametophyte. Fertilization occurs when sperm (n) and egg nuclei (n) unite.

8. Fertilization usually occurs more than a year after pollination. All eggs may be fertilized, but usually only one zygote develops into an embryo. The ovule becomes a seed, consisting of an embryo (new sporophyte) (2 n), food supply (gametophyte tissue) (n), and seed coat (2 n). Seeds are on the surface of ovulate scale and develop into a seedling and then mature sporophyte (2 n).

MAKE CONNECTIONS *What type of cell division occurs as a megaspore develops into a female gametophyte? Explain. (See*Figure 13.10.)**

*For suggested answer, see*Appendix A.**

Animation

Pine Life Cycle

image

The pine tree is the sporophyte; its sporangia are located on scalelike structures packed densely in cones. Like all seed plants, conifers are heterosporous. As such, they have two types of sporangia that produce two types of spores: microsporangia that produce microspores and megasporangia that produce megaspores. In conifers, the two types of spores are produced by separate cones: small pollen cones and large ovulate cones.

Pollen cones have a relatively simple structure: Their scales are modified leaves (microsporophylls) that bear microsporangia. Within each microsporangium, cells called microsporocytes undergo meiosis, producing haploid microspores. Each microspore develops into a pollen grain containing a male gametophyte. In conifers, the yellow pollen is released in large amounts and carried by the wind, dusting everything in its path.

Ovulate cones are more complex: Their scales are compound structures composed of both modified leaves (megasporophylls bearing megasporangia) and modified stem tissue. Within each megasporangium,

megasporocytes undergo meiosis and produce haploid megaspores inside the ovule. Surviving megaspores develop into female gametophytes, which are retained within the sporangia.

In most pine species, each tree has both types of cones. From the time pollen and ovulate cones appear on the tree, it takes nearly three years for the male and female gametophytes to be produced and brought together and for mature seeds to form from fertilized ovules. The scales of each ovulate cone then separate, and seeds are dispersed by the wind. A seed that lands in a suitable environment germinates, its embryo emerging as a pine seedling.

Early Seed Plants and the Rise of Gymnosperms

The origins of characteristics found in pines and other living seed plants date back to the late Devonian period (380 million years ago). Fossils from that time reveal that some plants had acquired features that are also present in seed plants, such as megaspores and microspores. For example, *Archaeopteris* was a heterosporous tree with a woody stem. It grew up to 20 m tall and had fernlike leaves. But it did not bear seeds and therefore is not classified as a seed plant.

The earliest evidence of seed plants comes from 360-million-year-old fossils of plants in the genus *Elkinsia***(Figure 30.5)**. These and other early seed plants lived 55 million years before the first fossils classified as gymnosperms and more than 200 million years before the first fossils of angiosperms. These early seed plants became extinct, and we don't know which extinct lineage gave rise to the gymnosperms.

Figure 30.5

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A fossil of the early seed plant *Elkinsia*.
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Photo of the fossilized seed plant Elkinsia, with the ovule labeled.

The oldest fossils of species from an extant lineage of gymnosperms are 305 million years old. These early gymnosperms lived in moist Carboniferous ecosystems that were dominated by lycophytes, horsetails, ferns, and other seedless vascular plants. As the Carboniferous period gave way to the Permian (299 to 252 million years ago), the climate became much drier. As a result, the lycophytes, horsetails, and ferns that dominated Carboniferous swamps were largely replaced by gymnosperms, which were better suited to the drier climate.

Gymnosperms thrived as the climate dried, in part because they have the key terrestrial adaptations found in all seed plants, such as seeds and pollen. In addition, some gymnosperms were particularly well suited to arid conditions because of the thick cuticles and relatively small surface areas of their needle-shaped leaves.

Gymnosperms dominated terrestrial ecosystems throughout much of the Mesozoic era, which lasted from 252 to 66 million years ago. In addition to serving as the food supply for giant herbivorous dinosaurs, these gymnosperms were involved in many other interactions with animals. Recent fossil discoveries, for example, show that some gymnosperms were pollinated by insects more than 100 million years ago—the earliest evidence of insect pollination in any plant group **(Figure 30.6)**. Late in the Mesozoic, angiosperms began to replace gymnosperms in some ecosystems.

Figure 30.6

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An ancient pollinator.
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Pollen grains on the back of a 110-million-year-old fossilized insect.

This 110-million-year-old fossil shows pollen on an insect, the thrip *Gymnopollisthrips minor*. Structural features of the pollen suggest that it was produced by gymnosperms (most likely by species related to extant ginkgos or cycads). Although most gymnosperms today are wind-pollinated, many cycads are insect-pollinated.

Gymnosperm Diversity

Although angiosperms now dominate most terrestrial ecosystems, gymnosperms remain an important part of Earth's flora. For example, vast regions in northern latitudes are covered by forests of conifers (see Figure 52.13).

Of the ten plant phyla (see Table 29.1), four are gymnosperms: Cycadophyta, Ginkgophyta, Gnetophyta, and Coniferophyta. It is uncertain how the four phyla of gymnosperms are related to each other. **Figure 30.7** surveys the diversity of extant gymnosperms.

Figure 30.7

Exploring Gymnosperm Diversity

Describes and illustrates examples of the following gymnosperm phyla: Cycadophyta, Ginkgophyta, Gnetophyta, and Coniferophyta.

Gymnosperm phyla are as follows.

Phylum Cycadophyta: The 300 species of living cycads have large cones and palm like leaves (true palm species are angiosperms). Unlike most seed plants, cycads have flagellated sperm, indicating their descent from seedless vascular plants that had motile sperm. Cycads thrived during the Mesozoic era, known as the

age of cycads as well as the age of dinosaurs. Today, however, cycads are the most endangered of all plant groups: 75% of their species are threatened by habitat destruction and other human actions.

Phylum Ginkgophyta: Ginkgo biloba is the only surviving species of this phylum; like cycads, ginkgos have flagellated sperm. Also known as the maidenhair tree, Ginkgo biloba has deciduous fanlike leaves that turn gold in autumn. It is a popular ornamental tree in cities because it tolerates air pollution well. Landscapers often plant only pollen-producing trees because the fleshy seeds smell rancid as they decay.

Phylum Gnetophyta: Phylum Gnetophyta includes plants in three genera: Gnetum, Ephedra, and Welwitschia. Some species are tropical, whereas others live in deserts. Although very different in appearance, the genera are grouped together based on molecular data. Gnetum. This genus includes about 35 species of tropical trees, shrubs, and vines, mainly native to Africa and Asia. Their leaves look similar to those of flowering plants, and their seeds look somewhat like fruits. Ephedra. This genus includes about 40 species that inhabit arid regions worldwide. These desert shrubs, commonly called Mormon tea, produce the compound ephedrine, which is used medicinally as a decongestant. Welwitschia. This genus consists of one species, Welwitschia mirabilis, a plant that can live for thousands of years and is found only in the deserts of southwestern Africa. Its strap like leaves are among the largest leaves known.

Phylum Coniferophyta, the largest gymnosperm phyla, consists of about 600 species of conifers (from the Latin conus, cone, and ferre, to carry), including many large trees. Most species have woody cones, but a few have fleshy cones. Some, such as pines, have needle-like leaves. Others, such as redwoods, have scale-like leaves. Some species dominate vast northern forests, whereas others are native to the Southern Hemisphere. Most conifers are evergreens; they retain their leaves throughout the year. Even during winter, a limited amount of photosynthesis occurs on sunny days. When spring comes, conifers already have fully developed leaves that can take advantage of the sunnier, warmer days. Some conifers, such as the dawn redwood, tamarack, and larch, are deciduous trees that lose leaves each autumn. Examples shown are Douglas fir, common juniper, European larch, Wollemi pine, sequoia, and bristlecone pine.

Douglas fir. This evergreen tree (Pseudotduga menziesii) provides more timber than any other North American tree species. Some uses include house framing, plywood, pulpwood for paper, railroad ties, and boxes and crates.

Common juniper. The berries of the common juniper (Juniperus communis) are actually ovule-producing cones consisting of fleshy sporophylls.

European larch. The needle-like leaves of this deciduous conifer (Larix decidua) turn yellow before they are shed in autumn. Native to the mountains of central Europe, including Switzerland's Matterhorn, depicted here, this species is extremely cold-tolerant, able to survive winter temperatures that plunge to negative 50 degrees Celsius.

Wollemi pine. Survivors of a conifer group once known only from fossils, living Wollemi pines (Wollemia nobilis) were discovered in 1994 in a national park near Sydney, Australia. At that time, the species consisted of 40 known trees. As a result of conservation efforts, it is now widely propagated. The inset photo compares the leaves of this living fossil with actual fossils.

Sequoia. This giant sequoia (Sequoiadendron giganteum) in California's Sequoia National Park weighs about 2,500 metric tons, equivalent to about 24 blue whales (the largest animals) or 40,000 people. The giant sequoia is one of the largest living organisms and also among the most ancient, with some individuals estimated to be between 1,800 and 2,700 years old. Their cousins, the coast redwoods (Sequoia sempervirens), grow to heights of more than 110 meters (taller than the Statue of Liberty) and are found only in a narrow coastal strip of northern California and southern Oregon.

Bristlecone pine. This species (Pinus longaeva), which is found in the White Mountains of California, includes some of the oldest living organisms, reaching ages of more than 4,600 years. One tree (not shown here) is called Methuselah because it may be the world's oldest living tree. To protect the tree, scientists keep its location a secret.

Concept Check 30.2

- Explain how the pine life cycle in Figure 30.4 reflects the five adaptations common to all seed plants.
- **VISUAL SKILLS** Based on Figure 30.4, compare and contrast the function of pollination and fertilization in sexual reproduction in pines.
- **MAKE CONNECTIONS** Early seed plants in genus *Elkinsia* are a sister group to a clade consisting of gymnosperms and angiosperms. Draw a phylogenetic tree of seed plants that shows *Elkinsia*, gymnosperms, and angiosperms; date the branch points on this tree using fossil evidence. (See Figure 26.5.)

*For suggested answers, see*Appendix A.**

Concept 30.3: The reproductive adaptations of angiosperms include flowers and fruits

On an evolutionary tree, a branch goes to gymnosperms and angiosperms; angiosperms are highlighted.

Branch point 1: First level-1 branch: Nonvascular plants (bryophytes). The second level-1 branch splits from branch point 2 into the first level-2 branch: seedless vascular plants. The second level-2 branch splits from

branch point 3 into the first level-3 branch: Gymnosperms and the second level-3 branch: Angiosperms.

Commonly known as flowering plants, angiosperms are seed plants with the reproductive structures called flowers and fruits. The name *angiosperm* (from the Greek *angion*, container) refers to seeds contained in fruits. Angiosperms are the most diverse and widespread of all plants, with more than 290,000 species (about 90% of all plant species).

Characteristics of Angiosperms

All angiosperms are classified in a single phylum, Anthophyta. Before considering the evolution of angiosperms, we will examine two of their key adaptations—flowers and fruits—and the roles of these structures in the angiosperm life cycle.

Flowers

The **flower** is a unique angiosperm structure that is specialized for sexual reproduction. In many angiosperm species, insects or other animals transfer pollen from one flower to the sex organs on another flower, which makes pollination more directed than the wind-dependent pollination of most species of gymnosperms. However, some angiosperms *are* wind-pollinated, particularly those species that occur in dense populations, such as grasses and tree species in temperate forests.

A flower is a specialized shoot that can have up to four types of modified leaves called floral organs: sepals, petals, stamens, and carpels **(Figure 30.8)**. Starting at the base of the flower are the **sepals**, which are usually green and enclose the flower before it opens (think of a rosebud). Interior to the sepals are the **petals**, which are brightly colored in most flowers and can aid in attracting pollinators. Flowers that are wind-pollinated, such as grasses, generally lack brightly colored parts. In all angiosperms, the sepals and petals are sterile floral organs, meaning that they do not produce sperm or eggs.

Figure 30.8

The structure of an idealized flower.

The structure of an idealized flower.

At the top of the stem is the receptacle. On this is the ovary, in which are the ovules. Sepals extend to either side of the receptacle. Around the ovary are petals, and within this are stamen and single carpel (simple pistil). Stamen have a filament and an anther. Single carpel refers to the ovary, the style from the ovary, and the stigma at the end of the style.

Within the petals are two types of fertile floral organs that produce spores, the stamens and carpels. Stamens and carpels are sporophylls, modified leaves that are specialized for reproduction. **Stamens** are

microsporophylls: They produce microspores that develop into pollen grains containing male gametophytes. A stamen consists of a stalk called the **filament** and a terminal sac, the **anther**, where pollen is produced. **Carpels** are megasporophylls: They produce megaspores that give rise to female gametophytes. The carpel is the "container" mentioned earlier in which seeds are enclosed; as such, it is a key structure that distinguishes angiosperms from gymnosperms. At the tip of the carpel is a sticky **stigma** that receives pollen. A **style** leads from the stigma to a structure at the base of the carpel, the **ovary**; the ovary contains one or more ovules. As in gymnosperms, each angiosperm ovule contains a female gametophyte. If fertilized, an ovule develops into a seed.

A flower may have one or more carpels. In many species, multiple carpels are fused into one structure. The term **pistil** is sometimes used to refer to a single carpel (a simple pistil) or two or more fused carpels (a compound pistil). Flowers also vary in symmetry **(Figure 30.9)** and other aspects of shape, as well as size, color, and odor. Much of this diversity results from adaptation to specific pollinators (see Figures 38.4 and 38.5).

Figure 30.9

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Flower symmetry.
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Compares radial symmetry (example is a daffodil) with bilateral symmetry (example is an orchid).

In radial symmetry, the sepals, petals, stamens, and carpels radiate out from a center. Any line through the central axis divides the flower into two equal parts. In bilateral symmetry, the flower can only be divided into two equal parts by a single line.

DRAW IT *Draw the single line that can divide the bilaterally symmetrical flower into two equal parts.*

*For suggested answer, see*Appendix A.**

Fruits

As seeds develop from ovules after fertilization, the ovary wall thickens and the ovary matures into a **fruit**. A pea pod is an example of a fruit, with seeds (mature ovules, the peas) encased in the ripened ovary (the pod).

Fruits protect seeds and aid in their dispersal. Mature fruits can be either fleshy or dry **(Figure 30.10)**. Tomatoes, plums, and grapes are examples of fleshy fruits, in which the wall (pericarp) of the ovary becomes soft during ripening. Dry fruits include beans, nuts, and grains. Some dry fruits split open at maturity to release seeds, whereas others remain closed. The dry, wind-dispersed fruits of grasses, harvested while on the plant, are major staple foods for humans. The cereal grains of maize, rice, wheat, and other grasses,

though easily mistaken for seeds, are each actually a fruit with a dry outer covering (the former wall of the ovary) that adheres to the seed coat of the seed within.

Figure 30.10

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Some variations in fruit structure.
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Compares the different structures of the fruits tomato, ruby grapefruit, nectarine, hazelnut, and milkweed.

Part a Tomato, a fleshy fruit with soft outer and inner layers of pericarp (fruit wall).

Part b Ruby grapefruit, a fleshy fruit with a firm outer layer and soft inner layer of pericarp.

Part c Nectarine, a fleshy fruit with a soft outer layer and hard inner layer (pit) of pericarp.

Part d Hazelnut, a dry fruit that remains closed at maturity.

Part (e) Milkweed, a dry fruit that splits open at maturity.

As shown in **Figure 30.11**, various adaptations of fruits and seeds help to disperse seeds (see also Figure 38.12). The seeds of some flowering plants, such as dandelions and maples, are contained within fruits that function like parachutes or propellers, adaptations that enhance dispersal by wind. Some fruits, such as coconuts, are adapted to dispersal by water. And the seeds of many angiosperms are carried by animals. Some angiosperms have fruits modified as burrs that cling to animal fur (or the clothes of humans). Others produce edible fruits, which are usually nutritious, sweet tasting, and vividly colored, advertising their ripeness. When an animal eats the fruit, it digests the fruit's fleshy part, but the tough seeds usually pass unharmed through the animal's digestive tract. When the animal defecates, it may deposit the seeds, along with a supply of natural fertilizer, many kilometers from where the fruit was eaten.

Figure 30.11

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Fruit adaptations that enhance seed dispersal.
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Compares adaptations that promote seed dispersal: explosive action, wings, seeds within berries, and barbs.

Part a Some plants such as flowers have mechanisms that disperse seeds by explosive action which releases the seeds.

Part b Wings enable maple fruits to be carried by the wind.

Part c Seeds within berries and other edible fruits are often dispersed in animal feces after they are eaten.

Part d The barbs of cockleburs facilitate seed dispersal by allowing the fruits to hitchhike on animals such as a dog.

Animation

Fruit Structure and Seed Dispersal

image

Play

image

Play

The Angiosperm Life Cycle

You can follow a typical angiosperm life cycle in **Figure 30.12**. The flower of the sporophyte produces microspores that form male gametophytes and megaspores that form female gametophytes. The male gametophytes are in the pollen grains, which develop within microsporangia in the anthers. Each male gametophyte has two haploid cells: a *generative cell* that divides, forming two sperm, and a *tube cell* that produces a pollen tube. Each ovule, which develops in the ovary, contains a female gametophyte, also known as an **embryo sac**. The embryo sac consists of only a few cells, one of which is the egg.

Figure 30.12

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The life cycle of an angiosperm.
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Life cycle of an angiosperm from the time when the megasporocyte in each ovule divides by meiosis through fertilization and when a seed germinates.

The cycle is as follows for a mature flower on sporophyte plant (2 n) with a carpel and anther.

1. In the Megasporangium (2 n) of each ovule, the megasporocyte divides by meiosis, producing four megaspores (n). One survives and gives rise to a female gametophyte (embryo sac which includes the egg (n), synergids next to the egg, polar nuclei in the central cell, and antipodal cells.
2. On the anther, each microsporangium contains microsporocytes (2 n) that divide by meiosis, producing microspores (n).

3. A microspore develops into a pollen grain. The generative cell of the male gametophyte (n) will divide, forming two sperm. The tube cell (with tube nucleus) will produce the pollen tube.
4. After pollination (transfer to the stigma), two sperm cells are discharged, traveling down the pollen tube inside the style. Here, the two sperm are seen in the pollen tube, first behind the tube nucleus, and then eventually discharged into each ovule.
5. Double fertilization occurs. One sperm fertilizes the egg, forming a zygote. The other sperm fertilizes the central cell, forming the endosperm (a food supply, 3n in this example).
6. The zygote develops into an embryo that is packaged along with food into a seed. (Fruit tissues surround the seed).
7. When a seed germinates, the embryo develops into a mature sporophyte.

VISUAL SKILLS *Based on this figure, what is the maximum number of seeds this flower could produce? To produce that number of seeds, at least how many pollen grains would have to germinate?*

*For suggested answer, see*Appendix A.**

Figure 30.12 Walkthrough

The life cycle of an angiosperm

Watch an explanation of this figure in the Figure 30.12 Walkthrough: The Life Cycle of an Angiosperm.

image

image

Animation

Angiosperm Life Cycle

image

Play

image

After its release from the anther, the pollen is carried to the sticky stigma at the tip of a carpel. Although some flowers self-pollinate, most have mechanisms that ensure **cross-pollination**, which in angiosperms is the transfer of pollen from an anther of a flower on one plant to the stigma of a flower on another plant of the

same species. Cross-pollination enhances genetic variability. In some species, stamens and carpels of a single flower may mature at different times, or they may be so arranged that self-pollination is unlikely.

The pollen grain absorbs water and germinates after it adheres to the stigma of a carpel. The tube cell produces a pollen tube that grows down within the style of the carpel. After reaching the ovary, the pollen tube penetrates through the **micropyle**, a pore in the integuments of the ovule, and discharges two sperm cells into the female gametophyte (embryo sac). One sperm fertilizes the egg, forming a diploid zygote. The other sperm fuses with the two nuclei in the large central cell of the female gametophyte, producing a triploid cell (3n). This type of **double fertilization**, in which one fertilization event produces a zygote and the other produces a triploid cell, is unique to angiosperms.

After double fertilization, the ovule matures into a seed. The zygote develops into a sporophyte embryo with a rudimentary root and one or two seed leaves called **cotyledons**. The triploid central cell of the female gametophyte develops into **endosperm**, tissue rich in starch and other food reserves that nourish the developing embryo.

What is the function of double fertilization in angiosperms? One hypothesis is that double fertilization synchronizes the development of food storage in the seed with the development of the embryo. If a particular flower is not pollinated or sperm cells are not discharged into the embryo sac, fertilization does not occur, and neither endosperm nor embryo forms. So perhaps double fertilization is an adaptation that prevents flowering plants from squandering nutrients on infertile ovules.

Another type of double fertilization occurs in some gymnosperm species belonging to the phylum Gnetophyta. However, double fertilization in these species gives rise to two embryos rather than to an embryo and endosperm.

As you read earlier, the seed consists of the embryo, the endosperm, and a seed coat derived from the integuments. An ovary develops into a fruit as its ovules become seeds. After being dispersed, a seed may germinate if environmental conditions are favorable. The coat ruptures and the embryo emerges as a seedling, using food stored in the endosperm and cotyledons until it can produce its own food by photosynthesis.

Angiosperm Evolution

Charles Darwin once referred to the origin of angiosperms as an “abominable mystery.” He was particularly troubled by the relatively sudden and geographically widespread appearance of angiosperms in the fossil record (about 100 million years ago, based on fossils known to Darwin). Recent fossil evidence and phylogenetic analyses have led to progress in solving Darwin’s mystery, but we still do not fully understand how angiosperms arose from earlier seed plants.

Fossil Angiosperms

Angiosperms are now thought to have originated in the early Cretaceous period, about 140 million years ago. By the mid-Cretaceous (100 million years ago), angiosperms began to dominate some terrestrial ecosystems. Landscapes changed dramatically as conifers and other gymnosperms gave way to flowering plants in many parts of the world. The Cretaceous ended 66 million years ago with mass extinctions of dinosaurs and many other animal groups and with further increases in the diversity and importance of angiosperms.

What evidence suggests that angiosperms arose 140 million years ago? First, although pollen grains are common in rocks from the Jurassic period (201 to 145 million years ago), none of these pollen fossils have features characteristic of angiosperms, suggesting that angiosperms may have originated after the Jurassic. Indeed, the earliest fossils with distinctive angiosperm features are of 130-million-year-old pollen grains discovered in China, Israel, and England. Early fossils of larger flowering plant structures include those of *Archaefructus***(Figure 30.13)** and *Leefructus*, both of which were discovered in China in rocks that are 125 million years old. Overall, early angiosperm fossils indicate that the group arose and began to diversify over a 20- to 30-million-year period—a less sudden event than was suggested by the fossils known during Darwin's lifetime.

Figure 30.13

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An early flowering plant.
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An early flowering plant.

Part a. Archaefructus sinensis, a 125-million-year- old fossil. This herbaceous species had simple flowers and bulbous structures that may have served as floats, suggesting it was aquatic. Recent phylogenetic analyses indicate that Archaefructus may belong to the water lily group. Scale: 5 centimeters.

Part b. Artist's reconstruction of Archaefructus sinensis shows tree-like structure with carpel at the tips, stamen along the branches, and possible floats below the stamen.

Can we infer traits of the angiosperm common ancestor from traits found in early fossil angiosperms? *Archaefructus*, for example, was herbaceous and had bulbous structures that may have served as floats, suggesting it was aquatic. But investigating whether the angiosperm common ancestor was herbaceous and aquatic also requires examining fossils of other seed plants thought to have been closely related to angiosperms. All of those plants were woody, indicating that the common ancestor was probably woody and probably not aquatic. As we'll see, this conclusion has been supported by recent phylogenetic analyses.

Angiosperm Phylogeny

Molecular and morphological evidence suggests that extant gymnosperm lineages had diverged from the lineage leading to angiosperms by 305 million years ago. Note that this does not imply that angiosperms originated 305 million years ago, but that the most recent common ancestor of extant gymnosperms and angiosperms lived at that time. Indeed, angiosperms may be more closely related to several extinct lineages of woody seed plants than they are to gymnosperms. One such lineage is the Bennettitales, an extinct group with flowerlike structures that may have been pollinated by insects **(Figure 30.14a)**. However, the Bennettitales and other similar lineages of extinct woody seed plants did not have carpels or flowers and hence are not classified as angiosperms.

Figure 30.14

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Angiosperm evolutionary history.
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Diagram of a longitudinal section through Bennettitales, and an evolutionary tree.

Part a. A close relative of the angiosperms? This reconstruction shows a longitudinal section through the flowerlike structures found in the Bennettitales, an extinct group of seed plants hypothesized to be more closely related to extant angiosperms than to extant gymnosperms.

Part b. Angiosperm phylogeny. This tree represents a current hypothesis of angiosperm evolutionary relationships, based on morphological and molecular evidence. Angiosperms originated about 140 million years ago. The dotted line indicates the uncertain position of the Bennettitales, a possible sister group to extant angiosperms.

Branch point 1: (roughly 300 million years ago (million years ago)). Living gymnosperms.

Branch point 2 (roughly 225 million years ago). Bennettitales (dotted line).

Branch Point 3 (roughly 145 million years ago). Amborella

Branch Point 4 (roughly 140 million years ago). Water lilies.

Branch Point 5 (roughly 135 million years ago). Star anise and relatives.

Branch Point 6 (roughly 130 million years ago). Magnoliids.

Branch Point 7 (roughly 125 million years ago). Monocots. Eudicots.

VISUAL SKILLS *Would the branching order of the phylogeny in (b) necessarily have to be redrawn if a*150-million-year-old*fossil monocot were discovered? Explain.*

*For suggested answer, see**Appendix A.*

Making sense of the origin of angiosperms also depends on working out the order in which angiosperm clades diverged from one another. Here, dramatic progress has been made in recent years. Molecular and morphological evidence suggests that the shrub *Amborella trichopoda*, water lilies, and star anise are living representatives of lineages that diverged from other angiosperms early in the history of the group **(Figure 30.14b)**. *Amborella* is woody, supporting the conclusion mentioned earlier that the angiosperm common ancestor was probably woody. Like the Bennettitales, *Amborella*, water lilies, and star anise lack ***vessel elements***, efficient water-conducting cells that are found in most present-day angiosperms. Overall, based on the features of ancestral species and angiosperms like *Amborella*, researchers have hypothesized that early angiosperms were woody shrubs that had small flowers and relatively simple water-conducting cells.

Evolutionary Links with Animals

Plants and animals have interacted for hundreds of millions of years, and those interactions have led to evolutionary change. For example, herbivores can reduce a plant's reproductive success by eating its roots, leaves, or seeds. As a result, if an effective defense against herbivores originates in a group of plants, those plants may be favored by natural selection—as will herbivores that overcome this new defense. Plant-pollinator and other mutually beneficial interactions also can have such reciprocal evolutionary effects.

Plant-pollinator interactions also may have affected the rates at which new species form. Consider the impact of a flower's symmetry (see Figure 30.9). On a flower with bilateral symmetry, an insect pollinator can obtain nectar (a sugary solution secreted by flower glands) only when landing in a certain position **(Figure 30.15)**. This constraint makes it more likely that pollen is placed on a part of the insect's body that will come into contact with the stigma of a flower of the same species. Such specificity of pollen transfer reduces gene flow between diverging populations and could lead to increased rates of speciation in plants with bilateral symmetry. This hypothesis can be tested using the approach illustrated in this diagram:

Evolutionary tree showing the common ancestor which branches into Bilateral clade and Radial clade, branch length represents time since divergence from a common ancestor. Number of clade species can be compared.

Figure 30.15

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A bee pollinating a bilaterally symmetrical flower.
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A honeybee lands inside the petals of the Scottish broom flower and the stamens make contact with the back of the bee.

To harvest nectar from this Scottish broom flower, a honeybee must land as shown. This releases a tripping mechanism that arches the flower's stamens over the bee and dusts it with pollen. Later, some of

this pollen may rub off onto the stigma of the next flower the bee visits.

Video

Bee Pollinating

image

image

A key step in this approach is to identify cases in which a clade with bilaterally symmetric flowers shares an immediate common ancestor with a clade whose members have radially symmetric flowers. One recent study identified 19 pairs of closely related “bilateral” and “radial” clades. On average, the clade with bilaterally symmetric flowers had nearly 2,400 more species than did the related clade with **radial symmetry**. This result suggests that flower shape can affect the rate at which new species form, perhaps by affecting the behavior of insect pollinators. Overall, plant-pollinator interactions may have contributed to the increasing dominance of flowering plants in the Cretaceous period, making angiosperms centrally important to ecological communities.

Angiosperm Diversity

From their humble beginnings in the Cretaceous period, angiosperms have diversified into more than 290,000 living species. Until the late 1990s, most systematists divided flowering plants into two groups, based partly on the number of cotyledons, or seed leaves, in the embryo. Species with one cotyledon were called **monocots**, and those with two were called **dicots**. Other features, such as flower and leaf structure, were also used to define the two groups. Recent DNA studies, however, indicate that the species traditionally called dicots are paraphyletic. The vast majority of species once categorized as dicots form a large clade, now known as **eudicots** (“true” dicots). **Figure 30.16** compares the main characteristics of monocots and eudicots. The rest of the former dicots are now grouped into four small lineages. Three of these lineages—*Amborella*, water lilies, and star anise and relatives—are informally called **basal angiosperms** because they diverged from other angiosperms early in the history of the group (see Figure 30.14b). A fourth lineage, the **magnoliids**, evolved later. **Figure 30.17** provides an overview of angiosperm diversity.

Figure 30.16

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Characteristics of monocots and eudicots.
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A table illustrates the differences in the following characteristics of monocots and dicots: embryos, leaf venation, stems, roots, pollen, and flowers.

blank	Embryos	Leaf venation	Stems	Roots	Pollen	Flowers
Monocot characteristics	One cotyledon	Veins usually parallel	Vascular tissue scattered	Root system usually fibrous (no main root)	Pollen grain with one opening	
Eudicot characteristics	Two cotyledons	Veins usually netlike	Vascular tissue usually arranged in ring	Taproot (main root) usually present	Pollen grain with three openings	Floral organs usually in multiples of four or five

Figure 30.17

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Exploring Angiosperm Diversity
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Exploring angiosperm diversity: photos and descriptions of basal angiosperms, magnoliids, monocots, and eudicots.

Descriptions are as follows.

Basal angiosperms: Surviving basal angiosperms consist of three lineages comprising only about 100 species. The first lineage to have diverged from other angiosperms is represented today by a single species, Amborella trichopoda (right). The other surviving lineages diverged later: a clade that includes water lilies and a clade consisting of the star anise and its relatives. Water lily (Nymphaea Rene Gerard). Species of water lilies are found in aquatic habitats throughout the world. Water lilies belong to a clade that diverged from other angiosperms early in the group's history. Amborella trichopoda. This small shrub, found only on the South Pacific island of New Caledonia, may be the sole survivor of a branch at the base of the angiosperm tree. Star anise (Illicium). This genus belongs to a third surviving lineage of basal angiosperms.

Magnoliids: Magnoliids consist of about 8,000 species, most notably magnolias, laurels, and black pepper plants. They include both woody and herbaceous species. Although they share some traits with basal angiosperms, such as a typically spiral rather than whorled arrangement of floral organs, magnoliids are more closely related to eudicots and monocots. Southern magnolia (Magnolia grandiflora). This member of

the magnolia family is a large tree. The variety of southern magnolia shown here, called Goliath, has flowers that measure up to about a foot across.

Monocots: About one-quarter of angiosperm species are monocots — about 70,000 species. Some of the largest groups are the orchids, grasses, and palms. Grasses include some of the most agriculturally important crops, such as maize, rice, and wheat. Examples pictured are orchid (Lemboglossum rossii); barley (Hordeum vulgare), a grass; and pygmy date palm (Phoenix roebelenii).

Eudicots: More than two-thirds of angiosperm species are eudicots — roughly 170,000 species. The largest group is the legume family, which includes such crops as peas and beans. Also important economically is the rose family, which includes many plants with ornamental flowers as well as some species with edible fruits, such as strawberry plants and apple and pear trees. Most of the familiar flowering trees are eudicots, such as oak, walnut, maple, willow, and birch. Examples pictured are snow pea (Pisum sativum), a legume; dog rose (Rosa canina), a wild rose; and pyrenean oak (Quercus pyrenaica).

Concept Check 30.3

- It is said that an oak is an acorn's way of making more acorns. Write an explanation that includes these terms: sporophyte, gametophyte, ovule, seed, ovary, and fruit.
- Compare and contrast a pine cone and a flower in terms of structure and function.
- **WHAT IF?** Do speciation rates in closely related clades of flowering plants show that flower shape is *correlated with* the rate at which new species form or that flower shape is *responsible for* this rate? Explain.

*For suggested answers, see*Appendix A.**

Concept 30.4: Human welfare depends on seed plants

In forests and on farms, seed plants are key sources of food, fuel, wood products, and medicine. Our reliance on them makes the preservation of plant diversity critical.

Products from Seed Plants

Most of our food comes from angiosperms. Just six crops—maize, rice, wheat, potatoes, cassava, and sweet potatoes—yield 80% of all the calories consumed by humans. We also depend on angiosperms to feed livestock: It takes 5–7 kg of grain to produce 1 kg of grain-fed beef.

Today's crops are the products of artificial selection—the result of plant domestication that began about 12,000 years ago. To appreciate the scale of this transformation, note how the number and size of seeds in domesticated plants are greater than those of their wild relatives, as in the case of maize and the grass teosinte (see Figure 38.16). Scientists can glean information about domestication by comparing the genes of crops with those of wild relatives. With maize, dramatic changes such as increased cob size and loss of the hard coating around teosinte kernels may have been initiated by as few as five mutations.

Flowering plants also provide other edible products. Two popular beverages come from tea leaves and coffee beans, and you can thank the cacao tree for cocoa and chocolate. Spices are derived from various plant parts, such as flowers (cloves, saffron), fruits and seeds (vanilla, black pepper, mustard), leaves (basil, mint, sage), and even bark (cinnamon).

Many seed plants are sources of wood, which is absent in all living seedless plants. Wood consists of tough-walled xylem cells (see Figure 35.22). It is the primary source of fuel for much of the world, and wood pulp, typically derived from conifers such as fir and pine, is used to make paper. Wood remains the most widely used construction material.

For centuries, humans have also depended on seed plants for medicines. Many cultures use herbal remedies, and scientists have extracted and identified medicinally active compounds from many of these plants and later synthesized them. Willow leaves and bark have long been used in pain-relieving remedies, including prescriptions by the Greek physician Hippocrates. In the 1800s, scientists traced the willow's medicinal property to the chemical salicin. A synthesized derivative, acetylsalicylic acid, is what we call aspirin. Plants are also a direct source of medicinal compounds **(Table 30.1)**. In the United States, about 25% of prescription drugs contain an active ingredient from plants, usually seed plants.

Table 30.1

Examples of Plant-Derived Medicines

A table titled Table 30.1 Examples of plant-derived medicines illustrates compounds, sources, and uses.

The table is as follows:

Table 30.1 Examples of Plant-Derived Medicines
Compound
Atropine
Digitalin
Menthol

Quinine
Taxol
Tubocurarine
Vinblastine

Threats to Plant Diversity

Although plants may be a renewable resource, plant diversity is not. The exploding human population and its demand for space and resources are threatening plant species across the globe. The problem is especially severe in the tropics, where more than two-thirds of the human population live and where population growth is fastest. About 63,000 km2(15 million acres) of **tropical rain forest** are cleared each year **(Figure 30.18)**, a rate that would completely eliminate the remaining 11 million km2 of tropical forests in 175 years. The loss of forests reduces the absorption of atmospheric carbon dioxide (CO_2) that occurs during photosynthesis, potentially contributing to global warming. Also, as forests disappear, so do large numbers of plant species. Of course, once a species becomes extinct, it can never return.

Figure 30.18

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Clear-cutting of tropical forests.
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One satellite image shows dense forest as a large region of green, and the other shows the same region with a variety of colors to indicate urban development.

Over the past several hundred years, nearly half of Earth's tropical forests have been cut down and converted to farmland and other uses. A satellite image from 1975 (left) shows a dense forest in Brazil. By 2012 (right image), much of this forest had been cut down. Deforested and urban areas are shown as light purple.

A two part satellite image of forests in Brazil, affected by urban development. The first image on the left shows a bright green region in 1975, with one branched region in brown. The second image on the right shows patches of green interspersed with large patches of brown. A large patch and small dots of purple indicate complete deforestation and urban development. Scale bar: 5 kilometers.

The loss of plant species is often accompanied by the loss of insects and other rain forest animals. Scientists estimate that if current rates of loss in the tropics and elsewhere continue, 50% or more of Earth's species will become extinct within the next few centuries. Such losses would constitute a global mass extinction,

rivaling the Permian and Cretaceous mass extinctions and forever changing the evolutionary history of plants (and many other organisms).

Many people have ethical concerns about contributing to the extinction of species. In addition, there are practical reasons to be concerned about the loss of plant diversity. So far, we have explored the potential uses of only a tiny fraction of the more than 325,000 known plant species. For example, almost all our food is based on the cultivation of only about two dozen species of seed plants. And fewer than 5,000 plant species have been studied as potential sources of medicines. The tropical rain forest may be a medicine chest of healing plants that could be extinct before we even know they exist. If we begin to view rain forests and other ecosystems as living treasures that can regenerate only slowly, we may learn to harvest their products at sustainable rates.

Concept Check 30.4

- Explain why plant diversity can be considered a nonrenewable resource.
- **WHAT IF?** How could phylogenies be used to help researchers search more efficiently for novel medicines derived from seed plants?

*For suggested answers, see*Appendix A.**